

LDA

AI-Based CAD Building Plan Verification & Pre-Scrutiny System

Faster, transparent and rule-based building plan scrutiny

**First CAD-Based AI Building Plan Verification System for
LDA / local authority use**

DG-level / CS-level review

CAD-based

DWG

DXF

PDF

OCR

Rules

GIS

Report

Executive Summary

The system reduces manual scrutiny friction while keeping officers in control of final decisions.

- Manual review is slowed by varied CAD/PDF submissions, repeated measurements and inconsistent drawing standards.
- AI-assisted pre-scrutiny turns drawings, documents, map evidence and rules into a structured report.
- The officer remains final authority, with every recommendation and override captured in an auditable trail.

Executive takeaway

A practical public-sector decision-support layer for real submissions, not a replacement for official review.

Strategic Positioning

LDA is building a locally practical CAD-based AI scrutiny model instead of forcing a high-cost BIM-only shift.

Developed-market approval models

BIM/Revit workflows are advanced and powerful

Require strict model standards and trained BIM ecosystem

Adoption depends on broad industry maturity

LDA CAD-based AI model

Works with existing architect submissions

Supports DWG, DXF, CAD and PDF

Fits current officer capacity and phased rollout

LDA is not simply copying high-cost BIM approval models from developed countries. It is building a practical CAD-based AI scrutiny model suited to local market realities.

Current Manual Process Challenges

Manual scrutiny remains document-heavy, interpretation-heavy and difficult to audit consistently.

Manual review

Repeated plan reading and document follow-up

Inconsistent interpretation

Different reviewers may check drawings differently

CAD layer issues

Missing or non-standard layer naming slows validation

Measurement checking

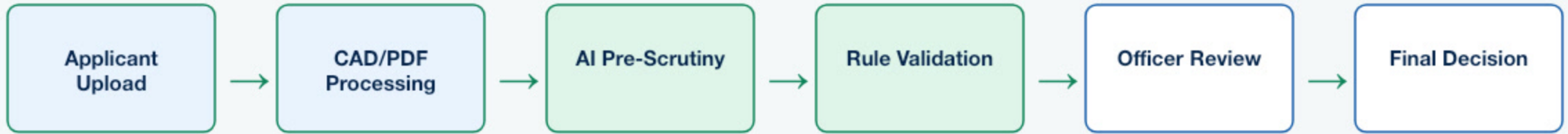
Setbacks, FAR and coverage require repetitive calculation

Limited audit trail

Evidence, remarks and decisions are hard to connect end to end

Proposed AI-Enabled Process

The workflow converts applicant submissions into structured officer-ready evidence.



Output: a structured scrutiny package for officer review, not an automated final approval.

Technology We Are Using

The platform combines CAD intelligence, GIS verification, rule logic and officer workflow controls.

Plan intelligence

CAD/DWG/DXF/PDF parsing

OCR/text extraction

Measurement detection

Decision support

Rule engine

ML confidence scoring

AI report generation

Public-sector workflow

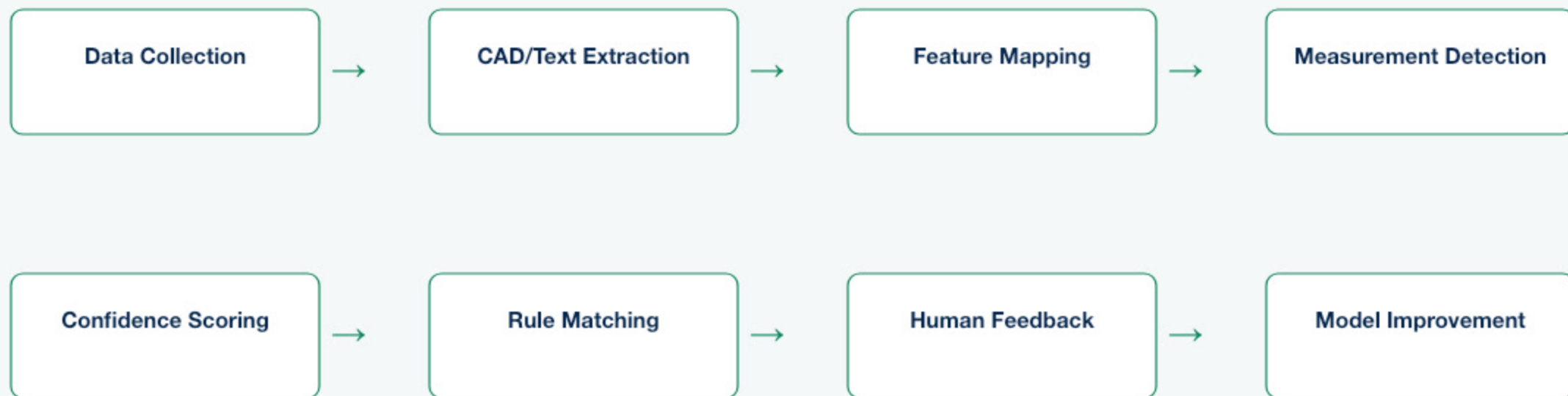
GIS/map verification

Officer dashboard

Audit logs

Machine Learning Process

The model improves through structured extraction, confidence scoring and officer feedback.



CAD-Based AI System vs BIM/Revit-Based Approval System

BIM/Revit approval is advanced; CAD-based AI is faster and more practical for local adoption.

Area	BIM/Revit-Based Approval System	LDA CAD-Based AI Scrutiny System
Input Format	BIM/Revit model required	CAD/DWG/DXF/PDF supported
Cost for Architects	High software and training cost	Works with existing CAD workflow
Adoption Barrier	Requires industry-wide BIM shift	Compatible with current submissions
Public Sector Readiness	Requires specialized BIM review teams	Supports existing officers with AI assistance
Implementation Speed	Slower due to ecosystem change	Faster pilot and phased rollout possible
Suitability	Highly digitized construction markets	Developing countries and local authorities
Scrutiny Method	Model-based validation	CAD/text/dimension/rule-based AI validation
Human Review	Still required for exceptions	Officer remains final decision authority

Feature Comparison

AI assistance standardizes repetitive checks and gives officers a clearer review package.

Capability	Manual System	AI-Assisted System
Plan reading	Manual visual review	CAD/text extraction with confidence
Measurement checking	Manual calculation	System-calculated values with officer review
Rule verification	Reviewer-led	Rule-wise pass/fail/manual review
Report generation	Manual notes	Structured AI scrutiny report
Decision support	Limited	Recommendation plus evidence
Audit trail	Fragmented	Logs, remarks and status history
Processing time	Longer and variable	Faster pre-scrutiny and triage

Why CAD-Based AI Is Right for Pakistan / Developing Countries

The approach works with the market as it exists today while keeping a path open for future BIM maturity.

Works with existing CAD drawings

Avoids costly BIM/Revit subscriptions

Faster adoption

Supports officers

Confidence-based scrutiny

Gradual future BIM transition

Lower training burden

Locally practical digital approval model

Key Benefits

The system improves speed, consistency and citizen-facing transparency.

Faster scrutiny

Pre-checks reduce repetitive officer workload

Reduced manual burden

Officers focus on exceptions and judgment

Transparent recommendations

Rule-wise evidence is visible

Better compliance

Common checks are standardized

Standardized reports

Consistent outputs across cases

Improved citizen service

Clearer status and faster response

Resource Benefits

AI pre-scrutiny lets the department scale review capacity without proportional staff growth.

Less repetitive checking

Measurements, layers and report checks move into pre-scrutiny.

Officers focus on exceptions

Human effort shifts to ambiguous, high-risk and policy-sensitive cases.

Reduced back-and-forth

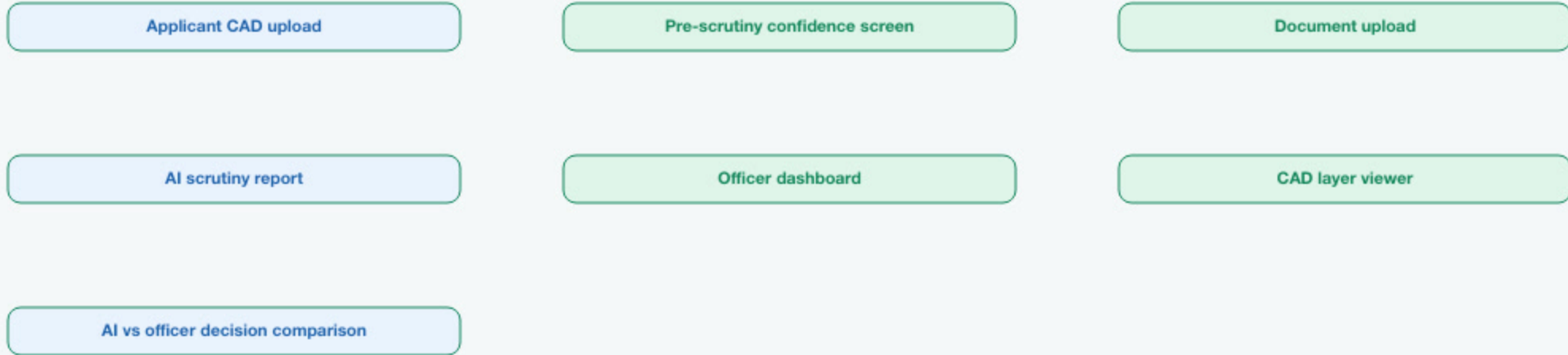
Applicant issues can be identified earlier with structured observations.

Better workload distribution

Cases can be triaged by confidence, risk and review status.

Where We Stand Today

A working implementation already covers the main public submission and officer review surfaces.



Screenshots / Demo Flow

Final review deck should use real product screenshots for each major workflow state.

SCREENSHOT PLACEHOLDER

Applicant upload screen

SCREENSHOT PLACEHOLDER

AI pre-scrutiny popup

SCREENSHOT PLACEHOLDER

Map verification

SCREENSHOT PLACEHOLDER

AI scrutiny report

SCREENSHOT PLACEHOLDER

AD ePermit dashboard

SCREENSHOT PLACEHOLDER

CAD viewer

AI Scrutiny Report Overview

The report is designed as officer-ready evidence, not a black-box AI answer.

Overall recommendation

Shown as structured report evidence for officer review.

Confidence score

Shown as structured report evidence for officer review.

CAD quality

Shown as structured report evidence for officer review.

Text layer checks

Shown as structured report evidence for officer review.

Extracted measurements

Shown as structured report evidence for officer review.

Rule-by-rule pass/fail/manual review

Shown as structured report evidence for officer review.

Strategic Advantage

LDA can establish a first-mover model for CAD-based AI scrutiny across local authorities.

First-mover position in CAD-based AI scrutiny

Lower adoption cost than BIM-only systems

Faster implementation in existing ePermit workflow

Better fit for local architects and officers

Scalable for other authorities

Future-ready for BIM/Revit integration later

Challenges

Known implementation risks are manageable through standards, feedback loops and phased rollout.

Missing CAD layers

Requires standards, training and feedback loop.

Absent text layers

Requires standards, training and feedback loop.

Different drawing standards

Requires standards, training and feedback loop.

Officer feedback requirement

Requires standards, training and feedback loop.

Integration dependencies

Requires standards, training and feedback loop.

Change management

Requires standards, training and feedback loop.

Training

Requires standards, training and feedback loop.

Steps of Implementation

A phased approach reduces operational risk while improving the rule engine with real feedback.



Governance & Human Review

The system assists decisions; it does not replace statutory authority.

AI gives recommendation only

Designed for transparent decision support.

Officer remains final authority

Statutory accountability remains with the reviewing authority.

Every decision is logged

Designed for transparent decision support.

Manual override allowed with remarks

Designed for transparent decision support.

Expected Impact

A practical CAD-based AI layer can improve speed, transparency and compliance outcomes.

Reduced scrutiny time

Earlier triage and automated report preparation

Improved transparency

Rule-wise evidence and confidence scores

Faster citizen response

Clearer application status and observations

Better compliance

Standardized rule checks

Lower workload

Less repetitive plan checking

Final Roadmap / Way Forward

The next workstream should move from working system to controlled pilot evidence.

Complete officer feedback module

Improve CAD layer selection UX

Add Google Places search where plot signal is missing

Strengthen AI confidence logic

Improve chatbot/rule assistant

Prepare pilot report

Towards Faster, Transparent & Intelligent Building Plan Approval

A locally optimized AI-assisted scrutiny framework built around real-world CAD submissions.

Thank you